

ELECTROMECH CORPORATION

RHCE / RHCSA / RHCSA Training
Training by experts



RHCSA

28th April 2011

RHCE Exam

28th April 2011

RHCSA

29th April 2011

RHCSA 423

29th April 2011

Ahmedabad, Baroda, Rajkot- Gujarat

**"Come home of
Open Source"**



TRAINING PARTNER

**Redhat Channel Partner
and Training Partner**

Ahmedabad:

404, Maulik Arcad, Nr. Mansi
cross Road, Satellite Rd

Baroda:

25, The Emperor, Fatehgunj

Ph.: 079 40027898/9327218003/
9327247499, Rajkot: 09427576240

[Http://www.electromech.info](http://www.electromech.info)

E-mail: electromech@electromech.info

failed, the stand-by server would be converted into a live server manually, ensuring a smooth flow of data. It also provided PITR (point-in-time recovery), which allowed data to be restored or recovered to the state it was in at any time since PITR logging was started for that database. However, this was not a mature solution, as it had to be worked on manually. There was a chance that the server could crash, or even go down due to a power failure.

The cherry on top: PostgreSQL 9.0

While the stand-by and *Memcached* server partly met the requirements of the website, it was PostgreSQL 9.0 that completely solved the problems associated with manual replication. In version 9.0, streaming replication is automated, and people do not have to manually convert the stand-by server into a live server. Singh explains, "We also felt the need for a load balancer, as we get high traffic on the website, and implemented Pgpool II. This distributes load uniformly across different database servers. For example, if Database Server 1 has too much load, and Database Server 2 is less loaded, the load balancer shifts some of the load from Server 1 to 2."

Ibnlive.in.com currently runs on five back-end servers: one *Memcached*, one load balancer and three database servers, and Singh believes this takes care of future requirements. "The replication feature has brought flexibility with it. If our Web traffic suddenly increases to 15 million visitors a day, we can deploy two more servers to go live, making it a much more scalable solution than what we previously used, i.e., MySQL. The server could not handle 2000-3000 connections

and would fail, but PostgreSQL now handles over 4000 connections smoothly. It also handles a good volume of data, as we serve about 32 GB of data from the database server on a daily basis. In terms of concurrency, the website is able to handle over a 100 connections at a time, enabling over 8 million page views in a day, as the *Memcached* server handles it. Also, from the application point of view, it does not get corrupted easily," Singh reveals.

The site had first migrated to PostgreSQL 8.0 from MySQL in 2007. "We first tested PostgreSQL on some of our servers. The ease of configuration and tuning made a huge difference, and we were convinced about it since viewers did not have to wait long to see their requested data," Singh explains. The team was satisfied with the migration to PostgreSQL 8.0, since the coders did not have to add, delete or modify much data. It was extracted in MySQL format and inserted in PostgreSQL, with minimal modification using PHP coding. The company quickly upgraded to 8.3 the next year, and later, to 9.0 in 2010.

Singh is planning to upgrade to PostgreSQL 9.1, but only after testing it on the servers. "It comes with stability, and also offers a synchronous solution for replication. Now, asynchronous solutions allow some delay between the time of a commit and its propagation to the other servers, with the possibility that some transactions might be lost in the switch to a backup server and those load-balanced servers might return slightly stale results. The synchronous server will replicate data written on the main master server to the other servers at the same time, making it possible to avoid delay," Singh concludes. 

By: Jalaja Ramanunni

The author is a journalist at the EFY Group. She is a fan of anything portable, and loves exploring different genres of music and movies.